PREPARED FOR CODE AND THEORY

Donald Richardson

Creative technology leader

About me

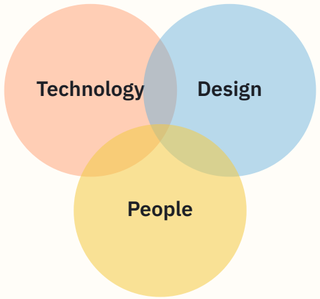


20+

years in creative technology and engineering

7+

years leading teams and projects



I build the systems behind creative work that fit how real people work.

Core skills

- System architecture and platform design
- Cloud, DevOps and web development
- Generative AI and workflow automation
- Creative technology and real-time media

PART ONE

Selected Work

Projects from Gensler and House of Apex.

Verizon Innovation Center, Boston

GENSLER · 2022 · FOR VERIZON

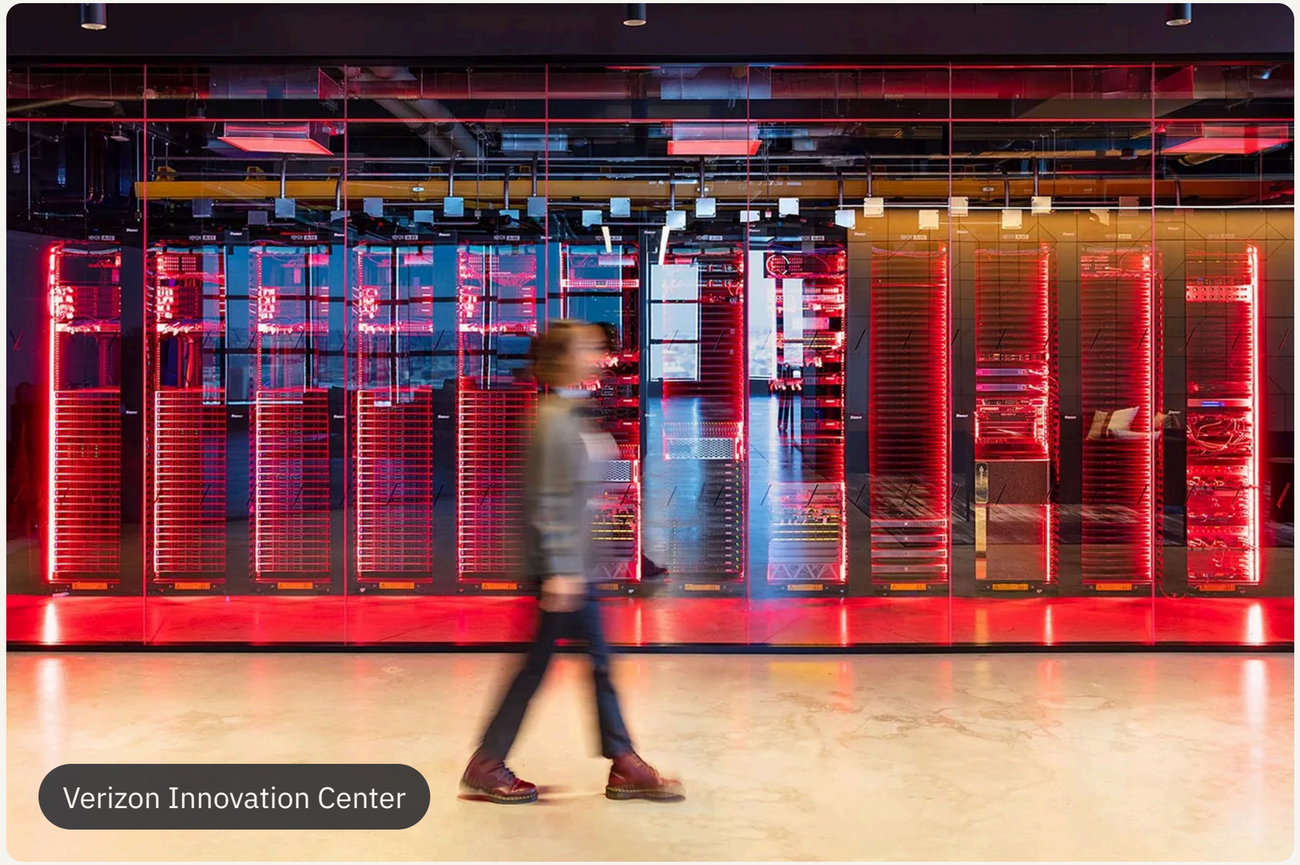
Built across three floors of Verizon’s new Boston headquarters, the center is where enterprise customers explore Verizon’s solutions and work with its experts to test their own ideas.

Visitors are greeted by an elevator lobby welcome moment, then embark on a journey built around them. Modular kiosk carts tell different demo stories, a portable immersive AR room transports visitors to Verizon labs in other cities, and transformable workshop spaces host hands-on sessions. A flip screen wall frames conversations with customers about their industry. Two immersive theaters accented with dynamic digital sculptures tell the bigger stories, and one hides a secret portal.

3
floors

17
touchpoints

1,200
story elements



Verizon Innovation Center

What I did

VERIZON · TECHNICAL LEAD AND ARCHITECT

THE SYSTEM

One platform behind every experience

- Content management and distribution for every touchpoint
- Events, sensors and show control tied together
- Hands-on with key pieces, from content sync to presence sensing

THE PROJECT

From pitch to installation

- Prototypes, technical design and value engineering
- Budget and schedule for the technology
- Design, AV, fabrication and outside developers working as one team

THE HANDOFF

Built into how they work

- Verizon’s IT, AV and content teams involved early
- Operations shaped around their existing workflows
- Process and troubleshooting documentation

Consulting firm HQ, Manhattan

GENSLER · 2021 · FOR A CONSULTING FIRM

A headquarters designed to bring employees back to the office, celebrating the firm’s impact on the world and its connection to New York City.

Dynamic communication, featured stories and artful activations run throughout. Visitors are greeted by a presence-driven welcome wall, and elevator lobby screens keep employees up to date on every floor. An interactive staircase installation animates as people climb, and a curved screen in the executive lobby grows a generative landscape as people walk by. An executive presentation room is set up for presenting.



18
floors

637,000
square feet

20+
screens

What I did

CONSULTING FIRM HQ · TECHNICAL LEAD AND ARCHITECT

THE SYSTEM

Content, distribution and events

- CMS and distribution for 20+ screens on a dozen floors
- Scheduled, templated internal communications
- Events behind the presence-driven experiences

THE PROJECT

Owning the technology end to end

- Budget, timeline and architecture
- Connecting design and other internal teams to the technology
- Coordination with hardware, AV, lighting, sound and other vendors

THE CLIENT

Built to the firm’s standards

- Security and data privacy review with their infosec and legal teams
- Content strategy for internal communications
- A system their teams could publish into

GSync

GENSLER · SELF-STARTED

Digital experience projects vary in scope, scale, technology, teams and vendors. These differences mean that projects are mostly one-offs, built from scratch every time. The result is duplicated effort, missed efficiencies and a growing maintenance burden.

So I built GSync, a shared platform for content management, distribution and infrastructure, designed to be reused across projects while allowing each experience to remain unique.

What it was

- Schema-driven content models and CMS
- A shared GraphQL content layer
- Content distribution, sync and scheduling
- Change notifications and event forwarding
- Client adapters for web, TouchDesigner and more

How it was built

- Funded by client projects
- Improved iteratively with each project
- Its proven success led to its adoption as a standard for projects across Gensler studios

The Platform

GENSLER · PLATFORM TECHNOLOGY DIRECTOR

GSync's track record led the firm to form a platform technology group around it, which I co-led. The goal was to make digital experiences turnkey, cheaper to deliver, hardened and easier to maintain.

Cloud

- Self-hosted on AWS, provisioned with Pulumi
- Multi-tenant Webiny CMS with access governance across clients
- Event-driven, with RabbitMQ carrying jobs and events to each site
- Containerized workers for video transcoding and image processing

On site

- Reusable client applications as a standard foundation
- Packaged Electron apps as an admin interface for clients
- Local logging, metrics, alerts and dashboards at each site

Delivery and operations

- Central observability across sites with ELK, Prometheus and Grafana
- Client apps built in CI, stored in an artifact repository
- Feature flags for real automated updates
- Planned: continuous deployment to test environments

House of Apex

2025 TO 2026 · INNOVATION DIRECTOR

A creative studio where I worked on two fronts: how we would offer AI and automation services to clients, and how our own team used AI tools and automation.

New services for clients

- Helped develop a strategy for adding AI and automation services to our portfolio, and a plan to grow them incrementally
- Custom tools for individual campaigns, from browser-rendered scenes to faster video production
- Full production pipelines, as a managed service or in the cloud

Inside the studio

- Trained our freelance team on AI tools, including Figma-integrated assistants, Adobe Firefly and online generation platforms
- Automated design and content production, including a tool to template and batch-render video projects
- AI and automation tools for business operations, from invoicing to knowledge management

PART TWO

A sample pilot

A discovery readout for Aurora, a fictional consumer audio brand. A way to show my thinking and start a conversation, not a proposal.

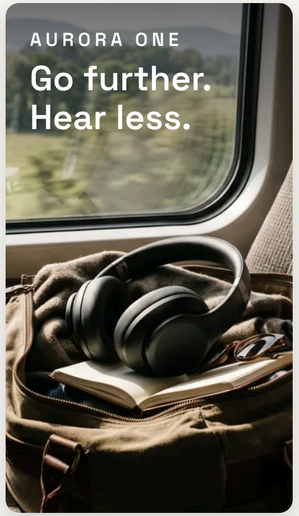
Meet Aurora

A consumer audio brand launching **Aurora One**, a wireless over-ear headphone, in three colorways.

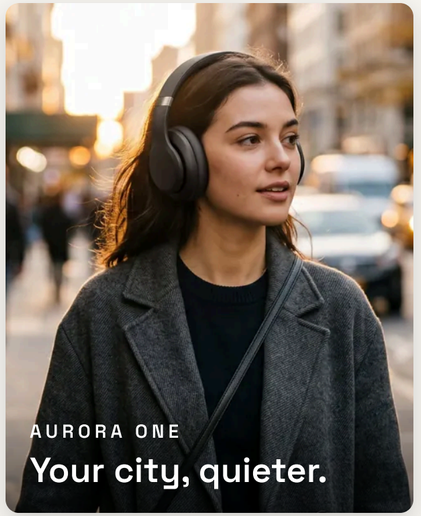


Midnight · Sand · Graphite

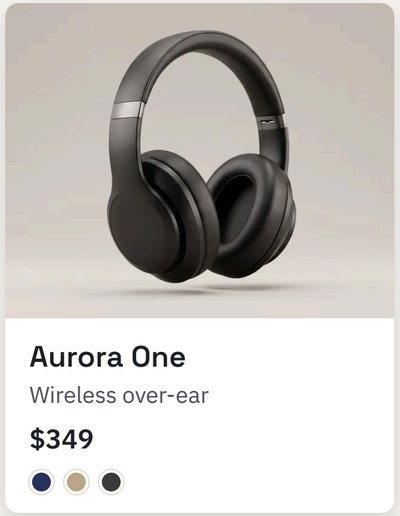
EXAMPLE LAUNCH ASSETS



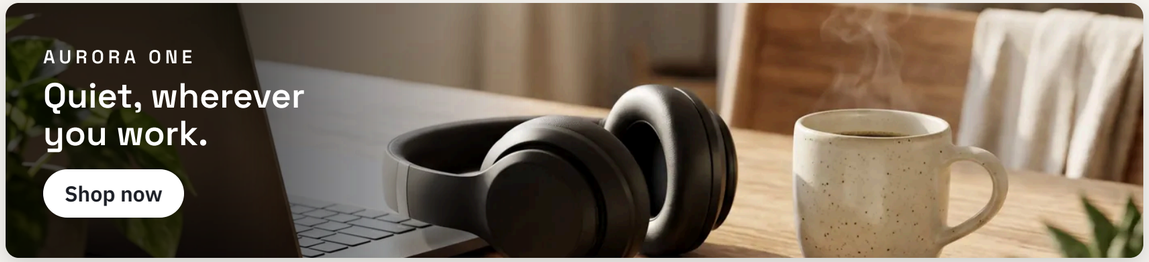
STORY · 9:16



SOCIAL POST · 4:5



RETAIL LISTING



WEB BANNER

Discovery phase

Simulated research

19 interviews, a survey with 41 responses, and three walkthroughs of how an asset gets made today, from brief to publish.

Who we talked to

6	**Marketing**	CMO, brand and channel leads
8	**In-house studio**	Designers, retouchers and producers
3	**Martech and IT**	Platform owners and security
2	**Legal**	Brand and IP counsel

WHAT THE SURVEY SHOWED

9 days
from brief to an approved image

~4
approval rounds per image

62%
of studio time spent resizing, versioning and retouching

1 in 3
requests is a resize or version of an existing image

> “We spend more time resizing than designing.”

Studio designer

What we heard in discovery

Simulated findings

MARKETING

Late launch imagery

Last launch, some colorways ran with only packshots in secondary channels until lifestyle images caught up weeks later.

SURVEY

Manual adaptation

One in three requests is a new size or version of an existing image, still built by hand in Photoshop.

IN-HOUSE STUDIO

A shoot for every colorway

Each color takes two weeks to shoot and retouch, and a late color change means doing it again.

PRODUCERS

Long approval cycles

Four rounds per image on average, with feedback spread across email, Slack and Workfront.

BRAND

Approved work is hard to find

Images aren’t tagged in AEM, so regional teams remake them, often with outdated logos and colors.

LEGAL

Unapproved AI tools

Designers have tried public image generators, and legal can’t tell what was used where.

Requirements

What the solution has to fit into, and the rules it has to follow.

Current stack

- Creative Cloud and AEM Assets
- Workfront for requests and approvals
- CAD from product engineering
- Product data in a PIM
- GenStudio for ads and market versions
- AWS as the company cloud

Guardrails

- Legal approves all generated imagery
- Indemnified models only
- Product shape, color and logo must be exact

The future state

People focus on

Art direction, campaign ideas, and judging what’s on brand.

The system handles

Rendering, scene variants, adaptation to every format, and a record of how every image was made.

1

Request

Marketing writes a brief in Workfront. Product data is pulled in from the PIM automatically.

2

Render the product

A 3D artist cleans up the converted CAD model once. Then every colorway renders automatically from each shot’s camera, with depth and masks.

3

Generate the scene

Art direction sets up each shot in 3D: camera, surface, product position and copy area. Firefly generates the scene from that layout.

4

Composite and adapt

The product layers straight onto the scene, already in place. It’s relit with contact shadows, then fitted to every format. Designers fix exceptions.

5

Review

The studio lead and legal approve templates up front, then review variants side by side in Workfront.

6

Deliver

Approved images go to AEM Assets with Content Credentials. Marketers add copy and build ads from them in GenStudio.

The pilot: six to eight weeks

3	6	5	90	4	~250
colorways	shots	formats	images	markets	deliverables

What the pilot covers

- Workfront, PIM and AEM integration
- A 3D master from Aurora’s CAD
- Six shots and the GenStudio templates, set up with Aurora’s studio
- 90 images, reviewed by the studio lead and legal
- Market versions by Aurora’s marketers in GenStudio
- An automated pipeline and process connecting all of the above

How we judge it

- Turnaround, brief to approved
- Cost per image, against today’s baseline
- Approval rounds and product accuracy
- How the studio team feels about it

Out of scope for the pilot: video, AR and more products · Run as a managed service, built to move into Aurora’s cloud later

How it runs

AURORA'S SYSTEMS

Workfront
Request and brief

PIM
Colorways, markets, copy, prices

3D master
Built once from engineering's CAD

PIPELINE ON AWS

EventBridge
Receives requests, starts runs

→

Step Functions
One job per variant, retries, waits for review

⇄

Review app
Pick scenes, adjust placement, fix exceptions

SQS queues to the workers

Convert and render
CAD conversion SDK, Blender on GPU instances (ECS)

Generate
Firefly Services API

Composite and adapt
Containers on Fargate, templates

Check
Code checks on Fargate, vision model on Bedrock

Shared data and monitoring

S3
Every file and intermediate

Aurora Postgres
Provenance record for every image

CloudWatch, X-Ray
Logs, alarms, tracing

Infrastructure as code, so it can be deployed into Aurora's own AWS account

BACK TO AURORA

AEM Assets
Approved images, tagged by SKU and colorway, with Content Credentials

Workfront
Approvals: shots up front, then every variant

GenStudio
Marketers add copy and build ads from approved images

The flow in detail

Pipeline · Hands-on · Approve in Workfront

1 Brief

- Marketing lead approves the brief
- ↓
- Webhook creates the launch, pulls PIM data, starts the workflow

2 3D master

- Converts CAD
- ↓
- 3D artist cleans up the model and sets materials per colorway
- ↓
- Validates against PIM dimensions
- ↓
- Studio lead approves against product photos

3 Shot setup

- Art direction sets camera, surface and copy area, and places the product. Designers build Express templates
- ↓
- Renders guide passes, generates previews
- ↓
- Studio lead approves each shot. Legal approves the rules once

4 First colorway

- Renders, generates and composites one colorway per shot
- ↓
- Designer picks the best generated scene
- ↓
- Studio lead approves before the full batch

5 Full batch

- Queues 90 jobs. Every image is checked automatically by code tests and a vision model
- ↓
- Designers fix flagged images in the variant PSD

6 Review

- Sends proofs to Workfront
- ↓
- Studio lead approves in bulk. Legal reviews every image in the pilot
- ↓
- Re-runs rejections, or creates a fix task
- ↓
- Designer fixes any images rejected for manual updates

7 Deliver

- Signs images, saves the record, uploads to AEM, closes the Workfront project
- ↓
- Marketers add copy and build ads in GenStudio

Adoption, audience by audience

Adoption starts with a small group in the pilot and widens at rollout.

Audience	In the pilot	At rollout
Studio designers	A champion and two or three designers review and art-direct the 90 images	Training for the whole studio, and designers own the templates
Marketing	Requests go through Workfront, and they see the turnaround	Every launch is requested this way
Legal	Approve the rules and templates, and review every image	Review by exception, backed by the record of each image
Martech and IT	Security review and the Workfront, PIM and AEM connections	Decide where it’s hosted, and take over the integrations

Where it pays off

The biggest wins first. The pilot measures each one against today’s baseline.

What we heard	Today	With the pipeline
A shoot for every colorway	Two weeks to shoot and retouch each color, and a late color change means doing it again	One 3D master. A new or changed color is a material change and a re-render
Late launch imagery	Some colorways ran with only packshots for weeks after launch	Every colorway has the full set of lifestyle images on launch day
Manual adaptation	62% of studio time goes to resizing, versioning and retouching	Every format comes from the shot setup. Designers fix exceptions
Long approval cycles	9 days and four rounds per image, with feedback spread across email, Slack and Workfront	Shots and rules approved once, then variants reviewed in batches in Workfront
Approved work is hard to find	Untagged images, remade by regional teams with outdated logos	Every image in AEM, tagged by SKU and colorway, ready for GenStudio
Unapproved AI tools	Legal can’t tell what was used where	An indemnified model, Content Credentials and a record behind every image

Aurora pilot · a fictional example